

Document Perturbation as Constrained Navigation on a Semantic Manifold

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Abstract

We introduce a framework for generating controlled document variations through constrained navigation in semantic space. Traditional document augmentation relies on synonym replacement, paraphrasing, or stochastic language-model sampling, providing little formal control over semantic drift. We propose a *semantic-budget framework* in which document transformations are perturbation operators acting on a semantic manifold $\mathcal{M}$. A document is a point in an embedding space; the text file is merely one coordinate representation of an underlying geometric object. Admissible transformations are constrained by a finite semantic distance budget, local curvature bounds, and conservation laws preserving core concepts. The framework is developed as exploration of the fibers of a projection $\pi : \mathcal{X} \rightarrow \mathcal{M}$, where $\mathcal{X}$ is the space of syntactically valid textual realizations and $\mathcal{M}$ encodes semantic identity. We introduce a semantic connection ∇ on the bundle that governs admissible transport, a curvature tensor Ω_P encoding the rigidity of different discourse communities, and a semantic accessibility entropy measuring the logarithmic volume of the reachable set. Motivated by colored-noise diffusion sampling [6], we prove that budget allocation weighted by a semantic completion field $\gamma(c, t)$ maximizes semantic diversity subject to a finite perturbation budget—constituting semantic colored noise. Mutation profiles are modelled as profile-dependent metric tensors M_P and connection structures ∇_P , enabling academic, legal, journalistic, technical, and literary registers to define semantic equivalence according to genuinely different geometric structures.

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1 Introduction

Controlled variation of natural-language documents underlies data augmentation for low-resource learning, robustness evaluation for retrieval and question-answering systems, deduplication benchmark construction, plagiarism detection calibration, and synthetic parallel corpus generation. Despite the centrality of these applications, the dominant approach remains conceptually simple: select words at random, substitute synonyms from a lexical database, and assume the resulting text remains meaningfully close to the original.

The limitations of this approach are well understood. WordNet-based substitution [19] treats all synonyms as equiprobable and ignores contextual fit, producing phrases such as *the investigator cautiously promulgated the constitution of the antique edifice* as a paraphrase of *the scientist carefully published the structure of the ancient building*. Large-language-model paraphrasing [22] resolves the contextual problem but introduces a different one: with no formal constraint on semantic drift, an unconstrained model may produce a document that preserves surface fluency while substantially altering meaning, logical structure, or domain-specific claims. Neither approach provides a principled account of *how much* the resulting document has changed or *which aspects* of the original have been preserved.

Most NLP papers treat the problem as follows: a document is a string; we transform the string; we measure whether the transformation was successful. The ontological status of the document is never questioned. We take a fundamentally different position. A document is a point on a semantic manifold; the text file is merely one coordinate representation of an underlying geometric object. Two different strings—*The scientist examined the structure* and *The researcher analyzed the architecture*—may occupy nearly the same region of the manifold. The geometric object is primary; the text is secondary.

This reframing elevates document perturbation from an NLP engineering task to a problem in constrained trajectory planning on a Riemannian manifold [1, 2]. The framework admits a natural formal language: fiber bundles, metric tensors, connections, holonomy, and completion-weighted energy allocation. The central question is no longer *which words should be replaced* but *which directions in semantic space are accessible from a given document under a given budget and a given set of conservation laws*.

We make the following contributions.

- (i) We introduce *semantic equivalence classes* via an approximate equivalence relation and show that fibers are exactly these classes, grounded in a locally linearized manifold structure (§3).
- (ii) We develop the *semantic fiber framework* (§4), defining semantic sections, reachable sets, and proving that controlled augmentation is sampling from a fiber at a specified radius.
- (iii) We introduce *semantic velocity* and *curvature-constrained trajectories* (§5), grounding curvature bounds in classical differential geometry.

- (iv) We introduce a *semantic connection* ∇ on the bundle (§5) and define admissible perturbations as horizontal lifts, resolving the non-holonomic failure modes of embedding-distance-only approaches.
- (v) We formalize conservation via *Noether-style semantic invariants* and *semantic gauge freedom* (§6), characterizing the admissible perturbation region as a fiber sub-manifold.
- (vi) We prove a *Budget Allocation Principle* for *semantic colored noise* (§7) and introduce *semantic accessibility entropy* with a monotonicity theorem.
- (vii) We model mutation profiles as *profile-dependent metric tensors and connection structures* (§8), deriving a *domain rigidity* measure from the ratio of connection curvature to admissible trajectory curvature.

2 Background

2.1 Document Augmentation

Synonym substitution using lexical resources such as WordNet [19] was among the first systematic approaches to document augmentation. Back-translation [20] introduced a noise-injection mechanism via round-tripping through an intermediate language. Large pre-trained language models [21, 22] enable fluent paraphrase generation but without quantitative control over semantic drift. Automatic evaluation metrics such as BERTScore [25] and BLEURT [26] provide post-hoc assessment but do not constrain generation.

2.2 Semantic Embedding Spaces

Dense document representations [14, 15, 16] map text into high-dimensional spaces where cosine similarity approximates semantic relatedness. A substantial literature [17, 8] has characterized the geometric structure of these spaces. Information geometry [3] provides a rigorous framework for treating such spaces as Riemannian manifolds equipped with the Fisher–Rao metric; our profile-dependent metric tensors are a discrete analogue.

2.3 Diffusion Processes and Colored Noise

Denoising diffusion probabilistic models [7] frame generation as iterative noise removal along a stochastic trajectory. The spectral bias of these processes—coarse structure resolves before fine detail [8, 9]—motivates *colored noise diffusion sampling* [6], which routes perturbation energy toward unresolved frequency bands via a completion field. Blue-noise approaches [11] and geometric analyses of sampling trajectories [10] further develop the idea that perturbation energy should respect existing structure.

2.4 Constraint-Based Optimization and Control Theory

Constrained text generation has been approached through lexical constraints [23], Lagrangian relaxation, and retrieval-augmented consistency enforcement. In control theory, constrained trajectory planning on smooth manifolds [1, 2] is a mature field; our budget constraints and curvature bounds are direct analogues. Optimal control theory [4, 5] provides the variational framework underlying our budget allocation principle. Gauge theories in physics [2] supply the connection and holonomy language adopted in §5 and §6.

3 The Semantic Manifold and Equivalence Classes

Let $\mathcal{X}$ denote the space of all syntactically valid documents in a given natural language. Under a fixed embedding function $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ we work in the embedded space $\phi(\mathcal{X})$.

Definition 3.1 (Semantic embedding). A *semantic embedding* is a map $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ such that $\|\phi(D)\| = 1$ for all $D \in \mathcal{X}$ and $\phi(D_1) \approx \phi(D_2)$ whenever D_1 and D_2 are judged semantically equivalent by human annotators.

Definition 3.2 (Semantic manifold). The *semantic manifold* $\mathcal{M} \subset \mathbb{R}^d$ is the image $\phi(\mathcal{X})$, endowed with the metric inherited from $\mathbb{R}^d$ restricted to $\mathcal{M}$.

3.1 Local Linearization

Although $\mathcal{M}$ may possess complicated global topology, we assume that every point $m \in \mathcal{M}$ admits a neighbourhood U_m locally diffeomorphic to $\mathbb{R}^k$ for some $k \leq d$ [1]. Consequently, local perturbation trajectories may be analyzed within the tangent space $T_m\mathcal{M}$, a k -dimensional real vector space. This justifies the use of local velocity vectors, curvature estimates, and metric tensors throughout the paper: all such objects are defined tangentially and then patched together across overlapping charts via standard manifold machinery [2].

3.2 Tangent Directions as Semantic Operations

A tangent vector $v \in T_m\mathcal{M}$ may be interpreted as an infinitesimal semantic operation: the direction in which the document’s meaning moves under an infinitesimal perturbation. Concretely, lexical substitutions induce directions that vary primarily within register and denotational axes; syntactic transformations perturb argument-structure and information-structure axes; discourse-level rewrites induce larger displacements spanning multiple semantic dimensions simultaneously. These qualitative distinctions motivate a decomposition of the tangent space into semantically distinguished subspaces:

$$T_m\mathcal{M} = T_m^{\text{lex}} \oplus T_m^{\text{syn}} \oplus T_m^{\text{sem}},$$

corresponding to lexical, syntactic, and semantic modes of variation. The decomposition need not be orthogonal, and the subspaces may overlap in practice; nevertheless, the three operator families introduced in §5 (lexical, syntactic, semantic) are designed to generate motion primarily within T_m^{lex} , T_m^{syn} , and T_m^{sem} respectively.

The semantic budget vector $\mathcal{B} = (B_\ell, B_s, B_p, \delta)$ therefore constrains not only the total magnitude of motion in $\mathcal{M}$ but also its decomposition across semantically distinct tangent directions. A budget that exhausts B_ℓ before reaching B_s or B_p has committed the perturbation to lexical variation and foreclosed the syntactic and semantic modes. This makes the budget vector a coarse-grained control over the *direction* of semantic travel, not merely its distance.

3.3 Semantic Equivalence Classes

Definition 3.3 (Approximate semantic equivalence). Two documents $D_1, D_2 \in \mathcal{X}$ are *semantically ε -equivalent*, written $D_1 \sim_\varepsilon D_2$, if $d_{\mathcal{M}}(\phi(D_1), \phi(D_2)) \leq \varepsilon$.

Proposition 3.4 ($\sim_\varepsilon$ is an approximate equivalence relation). *The relation $\sim_\varepsilon$ is reflexive and symmetric, and transitive up to 2ε : $D_1 \sim_\varepsilon D_2$ and $D_2 \sim_\varepsilon D_3$ imply $D_1 \sim_{2\varepsilon} D_3$.*

Proof. Reflexivity and symmetry are immediate. Transitivity up to 2ε follows from the triangle inequality: $d_{\mathcal{M}}(\phi(D_1), \phi(D_3)) \leq d_{\mathcal{M}}(\phi(D_1), \phi(D_2)) + d_{\mathcal{M}}(\phi(D_2), \phi(D_3)) \leq 2\varepsilon$. $\square$

The equivalence classes under $\sim_\varepsilon$ are the ε -neighbourhoods of each point on $\mathcal{M}$; documents in the same class are acceptable paraphrases of one another at tolerance ε .

Definition 3.5 (Semantic neighbourhood). $\mathcal{N}_r(D) = \{D' \in \mathcal{X} \mid d_{\mathcal{M}}(\phi(D), \phi(D')) \leq r\}$.

This establishes the logical chain: embeddings $\rightarrow$ manifold $\rightarrow$ local linearization $\rightarrow$ tangent decomposition $\rightarrow$ equivalence relation $\rightarrow$ neighbourhoods $\rightarrow$ fibers $\rightarrow$ perturbation.

4 Document Perturbation as Fiber Exploration

Definition 4.1 (Semantic projection). The *semantic projection* is the surjective map $\pi : \mathcal{X} \rightarrow \mathcal{M}$, $\pi(D) = \phi(D)$.

Definition 4.2 (Semantic fiber). The *semantic fiber* over $m \in \mathcal{M}$ is $\pi^{-1}(m) = \{D \in \mathcal{X} \mid \pi(D) = m\}$. The *approximate fiber* at tolerance ε is $\pi^{-1}(m)_\varepsilon = \{D \in \mathcal{X} : d_{\mathcal{M}}(\pi(D), m) \leq \varepsilon\}$.

Definition 4.3 (Semantic section). A *semantic section* is a map $s : \mathcal{M} \rightarrow \mathcal{X}$ such that $\pi \circ s = \text{id}_{\mathcal{M}}$. A section chooses one canonical textual realization for each semantic identity.

The original document D_0 is therefore not a privileged object but a *choice of section*: it selects one representative from the fiber $\pi^{-1}(\pi(D_0))$. Controlled augmentation explores other elements of the same fiber. Different choices of section over $\mathcal{M}$ —different canonical representatives—yield different augmentation starting points for the same underlying semantic content.

Proposition 4.4 (Fiber exploration as constrained augmentation). *Augmentation within semantic budget B is equivalent to sampling from $\pi^{-1}(\pi(D_0))_B$.*

Proof. By Definition 3.5, sampling within budget B is sampling from $\mathcal{N}_B(D_0)$. By Definition 4.2, the approximate fiber $\pi^{-1}(\pi(D_0))_B$ consists of all D' with $d_{\mathcal{M}}(\phi(D'), \phi(D_0)) \leq B$. Since $\pi = \phi$, these two sets coincide. $\square$

4.1 Reachable Sets

Definition 4.5 (Reachable set). The *reachable set* of D under budget B is $\mathcal{R}_B(D) = \{ D' \in \mathcal{X} \mid d_{\mathcal{M}}(\phi(D), \phi(D')) \leq B \}$.

A perturbation engine is more *expressive* if it can reach a larger fraction of $\mathcal{R}_B(D)$ [4]. By Proposition 4.4, $\mathcal{R}_B(D_0) = \pi^{-1}(\pi(D_0))_B$: the reachable set is the approximate fiber. Budget and fiber radius are the same quantity measured from different perspectives.

Remark 4.6 (Why synonym replacement can fail). A sequence of individually small steps can exit $\mathcal{R}_B(D_0)$ if the trajectory curves sharply. Two steps of size δ in perpendicular directions reach distance $\delta\sqrt{2}$ from the origin; two steps in opposite directions return to the origin. Bounding total displacement alone does not prevent this.

5 Semantic Distance Budgets, Velocity, and the Semantic Connection

Definition 5.1 (Perturbation operator). A *perturbation operator* $\mathcal{O} : \mathcal{X} \rightarrow \mathcal{X}$ maps a document to a modified version. Operators are organized by scale:

- (i) *Lexical operators*: synonym substitution, register shifts. Typical cost $\Delta \approx 0.001$ – 0.012 .
- (ii) *Syntactic operators*: active–passive, clause reordering. Typical cost $\Delta \approx 0.008$ – 0.020 .
- (iii) *Semantic operators*: LLM paraphrase, summarization–expansion. Typical cost $\Delta \approx 0.015$ – 0.040 .

The *cost* of applying $\mathcal{O}$ to D is $c(\mathcal{O}, D) = d_{\mathcal{M}}(\phi(D), \phi(\mathcal{O}(D)))$.

Definition 5.2 (Semantic budget vector). A *semantic budget vector* is $\mathcal{B} = (B_\ell, B_s, B_p, \delta) \in \mathbb{R}_{>0}^4$ specifying lexical, syntactic, and semantic budgets and a maximum step size $\delta > 0$. The total budget is $B = B_\ell + B_s + B_p$.

Definition 5.3 (Admissible perturbation sequence). A sequence of operators producing trajectory $D_0 \rightarrow D_1 \rightarrow \dots \rightarrow D_n$ is *admissible* with respect to $\mathcal{B}$ if:

$$\begin{aligned} d_{\mathcal{M}}(\phi(D_0), \phi(D_n)) &\leq B \quad (\text{total displacement}), \\ d_{\mathcal{M}}(\phi(D_{k-1}), \phi(D_k)) &\leq \delta \quad \forall k \quad (\text{step bound}). \end{aligned}$$

The perturbation problem is:

$$\max_{(\mathcal{O}_1, \dots, \mathcal{O}_n) \text{ admissible}} V(D_n) \quad (1)$$

where $V : \mathcal{X} \rightarrow \mathbb{R}$ is a diversity functional, typically $V(D') = d_{\mathcal{M}}(\phi(D_0), \phi(D'))$.

5.1 Semantic Velocity and Curvature

Definition 5.4 (Semantic velocity). The *semantic velocity* at step k is $v_k = \phi(D_k) - \phi(D_{k-1}) \in T_{\phi(D_{k-1})}\mathcal{M} \subset \mathbb{R}^d$.

Definition 5.5 (Discrete trajectory curvature). The *discrete curvature* at step k is

$$\kappa_k = 1 - \frac{\langle v_k, v_{k+1} \rangle}{\|v_k\| \|v_{k+1}\|}.$$

$\kappa_k = 0$ is straight-line motion; $\kappa_k = 2$ is reversal.

A *curvature-constrained trajectory* satisfies $\kappa_k \leq \kappa_{\max}$ at all interior steps.

Proposition 5.6 (Curvature reduces effective reachable radius). For $n = B/\delta$ maximum-size steps, bounding $\kappa_k \leq \kappa_{\max} < 1$ reduces maximum possible displacement below the unconstrained bound $\sqrt{n} \delta$.

Proof sketch. Unconstrained orthogonal steps yield $\|\sum_k v_k\| \leq \sqrt{n} \delta$ by Cauchy-Schwarz. With $\kappa_k \leq \kappa_{\max}$, consecutive velocities lie within a cone of half-angle $\theta = \arccos(1 - \kappa_{\max})$. By induction, the n -step displacement is bounded by $\delta \sum_{k=0}^{n-1} \cos^k \theta < \sqrt{n} \delta$ for $\kappa_{\max} < 1$. $\square$

5.2 Parallel Transport of Meaning and the Semantic Connection

The distance metric g specifies the cost of movement but does not specify how semantic quantities are transported along a trajectory. Certain semantically catastrophic mutations—negation reversal, causal inversion—are locally distance-preserving in embedding space while producing globally inadmissible documents. To address this failure mode we introduce a connection on the semantic bundle.

Definition 5.7 (Semantic connection). A *semantic connection* ∇ on the bundle $\pi : \mathcal{X} \rightarrow \mathcal{M}$ is a rule specifying, for each trajectory $D_0 \rightarrow D_1 \rightarrow \dots \rightarrow D_n$, how semantic quantities are transported from fiber to fiber. A quantity Q is *parallel transported* along the trajectory when

$$\nabla_{\dot{D}} Q = 0.$$

Admissible perturbations correspond to *horizontal trajectories* of the bundle: trajectories whose velocity lies entirely in the horizontal distribution $H \subset T\mathcal{X}$ defined by ∇ . Inadmissible semantic alterations correspond to motion with a nonzero vertical component, moving between fibers rather than within them.

5.3 Holonomy and Accumulated Drift

A perturbation trajectory may return to a neighbourhood of its starting point while failing to preserve all transported semantic quantities. Let Γ be a closed perturbation loop in $\mathcal{M}$ —a sequence of operators that returns the document’s embedding to its original position. Parallel transport around Γ via ∇ induces a *holonomy operator*

$$\text{Hol}_\Gamma : \pi^{-1}(\pi(D_0)) \rightarrow \pi^{-1}(\pi(D_0)).$$

If $\text{Hol}_\Gamma \neq \text{id}$, then even a closed embedding trajectory accumulates semantic drift: the document returns to the same embedding position but ends in a different semantic orbit, with some conserved quantity Q shifted by the holonomy.

Holonomy is nonzero precisely when the curvature $\Omega = d\nabla + \nabla \wedge \nabla$ of the connection is nontrivial over the region enclosed by Γ . This provides a direct operational interpretation of the curvature tensor: Ω_P measures how readily long perturbation chains accumulate invisible semantic drift even when each individual step is admissible. A high-curvature connection means that closed loops are dangerous; a flat connection means that order of operations is irrelevant and semantic drift cannot accumulate. The domain rigidity measure $R(P)$ defined in §8 formalizes the ratio of how much holonomy the connection can accumulate to how much the discourse community will tolerate.

Example 5.8 (Negation as a vertical direction). Introduce a semantic polarity coordinate $P(D) \in \{+1, -1\}$ where $+1$ denotes affirmation and -1 denotes negation. Define ∇ so that P is a horizontally conserved quantity: $\nabla_D P = 0$ along any admissible trajectory. A mutation changing P is not merely expensive; it moves in a vertical direction and exits the horizontal distribution entirely. The trajectory is inadmissible regardless of its embedding-space cost.

Example 5.9 (Causality as graph conservation). Let $G_D = (V, E)$ be the directed causal graph of document D , where nodes are propositions and edges encode causal or inferential dependencies. Define ∇ so that G_D is horizontally conserved: admissible perturbations must preserve G_D up to graph isomorphism. The documents A caused B and B caused A are not connected by any horizontal path in the bundle, even though their embedding vectors may be close, because any path between them must pass through a vertical displacement that inverts the edge direction.

The connection generalizes the projection-based conservation constraints of §6 by making conservation *path-dependent* rather than purely position-dependent. A quantity conserved via the connection remains invariant under parallel transport along any admissible trajectory; a position-based constraint merely requires that the quantity be the same at the endpoint. This is precisely the distinction between holonomic and non-holonomic constraints in classical mechanics: the current conservation formalism is holonomic (positions constrained), while the connection is non-holonomic (velocities/directions constrained).

6 Conservation Laws

Definition 6.1 (Protected concept set). A *protected concept set* is $\mathcal{C} = \{c_1, \dots, c_k\}$, each c_i represented by $\phi(c_i) \in \mathbb{R}^d$.

Definition 6.2 (Conservation constraint). A perturbation $D \rightarrow D'$ satisfies the *conservation constraint* with respect to $\mathcal{C}$ and tolerance $\varepsilon > 0$ if

$$|\langle \phi(D'), \phi(c_i) \rangle - \langle \phi(D), \phi(c_i) \rangle| \leq \varepsilon \quad \forall c_i \in \mathcal{C}.$$

6.1 Noether-Style Semantic Invariants

Definition 6.3 (Semantic conserved quantity). A function $Q : \mathcal{X} \rightarrow \mathbb{R}$ is a *semantic conserved quantity* if every admissible perturbation $D \rightarrow D'$ satisfies $|Q(D') - Q(D)| \leq \varepsilon$ for some declared tolerance ε .

Example 6.4 (Conserved charges in practice). (i) *Thesis projection*: $Q(D) = \langle \phi(D), \phi(\text{central claim}) \rangle$.

(ii) *Terminology*: $Q(D) = \langle \phi(D), \phi(t) \rangle$ for each domain term $t \in \{\text{RSVP, TARTAN, Friedmann, } \dots\}$.

(iii) *Citation preservation*: fraction of source citations appearing in D .

(iv) *Numerical claim conservation*: set of quantitative assertions in D , measured by entailment score.

(v) *Polarity*: $Q(D) = P(D) \in \{+1, -1\}$, conserved via ∇ .

(vi) *Causal graph*: $Q(D) = G_D$ up to isomorphism, conserved via ∇ .

Each conserved quantity defines a constraint hyper-surface in $\mathcal{M}$.

Proposition 6.5 (Invariant fiber sub-manifold). Let $Q_1, \dots, Q_n$ be smooth semantic conserved quantities with values $q_i = Q_i(D_0)$. Then the admissible perturbation space

$$\mathcal{F}_Q = \{ D \in \mathcal{X} \mid Q_i(D) = q_i, i = 1, \dots, n \}$$

forms a sub-manifold of the semantic fiber $\pi^{-1}(\pi(D_0))$, provided the gradients ∇Q_i are linearly independent on $\mathcal{F}_Q$.

Proof. By the regular level set theorem [1], the preimage of a regular value of a smooth map $Q = (Q_1, \dots, Q_n) : \mathcal{X} \rightarrow \mathbb{R}^n$ is a smooth sub-manifold of $\mathcal{X}$ of codimension n . Since $\mathcal{F}_Q \subset \pi^{-1}(\pi(D_0))$ by construction, it is a sub-manifold of the fiber. $\square$

Conservation is therefore not merely a constraint; it is geometry. The perturbation engine explores a lower-dimensional sub-manifold of the fiber, one whose dimension is reduced by one for each independent conserved quantity.

6.2 Semantic Gauge Freedom

Not every modification represents genuine semantic change. Certain surface variations alter textual realization while preserving all conserved quantities.

Definition 6.6 (Semantic gauge transformation). A *semantic gauge transformation* is a map $g : \mathcal{X} \rightarrow \mathcal{X}$ such that $Q_i(g(D)) = Q_i(D)$ for all conserved quantities Q_i and all $D \in \mathcal{X}$. Documents related by a gauge transformation are *gauge equivalent*.

Example 6.7 (Gauge-equivalent substitutions). The substitutions *researcher* $\leftrightarrow$ *scientist* and *investigate* $\leftrightarrow$ *analyze* are gauge transformations: they alter surface realization without moving the document’s projection onto thesis, polarity, or causal graph axes.

The perturbation problem therefore has two levels: finding gauge-equivalent realizations (cheap, within the same orbit of the gauge group), and genuinely exploring the fiber by moving between gauge orbits (more expensive, consumes semantic budget). The semantic budget B governs inter-orbit travel; gauge transformations are free with respect to the budget.

7 Semantic Colored Noise and Accessibility Entropy

Standard document perturbation is implicitly *semantic white noise*: it allocates perturbation energy uniformly across all semantic components, regardless of how much each has already been varied. We show that this is suboptimal and introduce a completion-weighted allocation principle motivated by colored noise diffusion sampling [6].

7.1 Semantic Completion Fields

Let $\mathcal{C}_{\text{sem}} = \{c_1, \dots, c_m\}$ be a decomposition of the document’s semantic content into components, concretely: {thesis, evidence, terminology, style, narrative, examples}.

Definition 7.1 (Semantic completion field). The *semantic completion field* is a function $\gamma : \mathcal{C}_{\text{sem}} \times [0, 1] \rightarrow [0, 1]$ where $\gamma(c, t)$ measures the fraction of variation budget already spent on component c by perturbation step t .

Example 7.2 (Completion landscape of a mature document). A document with a well-developed thesis and established terminology but relatively unvaried style and few examples might exhibit: $\gamma(\text{thesis}) = 0.95$, $\gamma(\text{terminology}) = 0.92$, $\gamma(\text{style}) = 0.40$, $\gamma(\text{examples}) = 0.25$. A uniform perturbation engine wastes budget on thesis and terminology; a completion-weighted engine routes variation toward style and examples.

7.2 Semantic Utility and the Budget Allocation Principle

Definition 7.3 (Semantic utility function). The *semantic utility* of allocating perturbation energy E to component c with current completion $\gamma(c)$ is

$$U_c(E) = \sqrt{E(1 - \gamma(c))},$$

representing diminishing returns in highly completed semantic regions: additional budget applied to a saturated component yields progressively less new variation.

Definition 7.4 (Semantic colored noise). A perturbation strategy produces *semantic colored noise* if energy allocated to component c at step t is

$$E(c, t) = \mathcal{B} \cdot \frac{1 - \gamma(c, t)}{\sum_j (1 - \gamma(j, t))}.$$

Uniform allocation $E(c, t) = \mathcal{B}/m$ is *semantic white noise*.

Theorem 7.5 (Budget Allocation Principle). Let $V(D') = \sum_c \sigma_c \cdot U_c(E(c))$ be the total semantic utility with weights $\sigma_c > 0$ and utility U_c as in Definition 7.3. Under the constraint $\sum_c E(c) = \mathcal{B}$ with $E(c) \geq 0$, semantic colored noise allocation maximizes V over all budget distributions.

Proof. Maximize $\sum_c \sigma_c \sqrt{E(c)(1 - \gamma(c))}$ subject to $\sum_c E(c) = \mathcal{B}$. By the method of Lagrange multipliers, at the optimum $\frac{\partial}{\partial E(c)} [\sigma_c \sqrt{E(c)(1 - \gamma(c))} - \lambda E(c)] = 0$, giving $\sigma_c \sqrt{(1 - \gamma(c))/(2\sqrt{E^*(c)})} = \lambda$, hence $E^*(c) \propto \sigma_c^2 (1 - \gamma(c))$. With uniform weights $\sigma_c = 1$ this yields $E^*(c) \propto 1 - \gamma(c)$, which is exactly the colored noise allocation of Definition 7.4. $\square$

Remark 7.6 (Relation to CNS). Davidson et al. [6] route diffusion energy toward $\sqrt{1 - \gamma(f, t)}$ over Fourier frequency bands. Theorem 7.5 establishes that the same energy-routing principle is optimal in the semantic domain when frequency bands are replaced by semantic components. The mathematical structure is identical; what changes is the space over which completion is measured.

Remark 7.7 (Semantic spectral bias). The CNS paper [6] identifies a *spectral bias* in diffusion models: coarse structure resolves before fine structure. The semantic analogue is *conceptual bias*: core thesis and terminology are established early and become increasingly rigid, while style, examples, and phrasing remain underexplored. Theorem 7.5 shows that completion-weighted allocation is not merely heuristically appealing but provably optimal under the utility model of Definition 7.3.

7.3 Semantic Accessibility Entropy

Definition 7.8 (Semantic accessibility entropy). The *semantic accessibility entropy* of a document D under budget B and active profile metric g_P is

$$S(D, B) = \log \text{Vol}_P(\mathcal{R}_B(D)),$$

where Vol_P denotes volume measured under the profile metric.

This quantity measures the logarithmic volume of semantic space accessible from D while respecting all budget and conservation constraints. Increasing budget increases semantic accessibility; conservation laws reduce it by restricting the admissible region to the sub-manifold $\mathcal{F}_Q \subset \pi^{-1}(\pi(D))$.

Proposition 7.9 (Monotonicity of semantic accessibility). *For $B_1 < B_2$, we have $S(D, B_1) \leq S(D, B_2)$.*

Proof. Since $B_1 < B_2$ implies $\mathcal{R}_{B_1}(D) \subseteq \mathcal{R}_{B_2}(D)$, their volumes satisfy $\text{Vol}_P(\mathcal{R}_{B_1}(D)) \leq \text{Vol}_P(\mathcal{R}_{B_2}(D))$, and taking logarithms preserves the inequality. $\square$

7.4 Entropy Reduction by Conservation

Conservation laws do not merely filter out inadmissible perturbations after the fact; they geometrically reduce the volume of the accessible region and therefore strictly decrease semantic accessibility entropy.

Proposition 7.10 (Conservation reduces accessibility). *Let $\mathcal{F}_Q \subseteq \mathcal{R}_B(D)$ be the admissible region determined by a collection of conserved quantities $Q_1, \dots, Q_n$ as in Proposition 6.5. Define the constrained accessibility entropy $S_Q(D, B) = \log \text{Vol}_P(\mathcal{F}_Q)$. Then*

$$\text{Vol}_P(\mathcal{F}_Q) \leq \text{Vol}_P(\mathcal{R}_B(D)),$$

and consequently $S_Q(D, B) \leq S(D, B)$.

Proof. Since $\mathcal{F}_Q \subseteq \mathcal{R}_B(D)$ by construction, monotonicity of Vol_P under set inclusion gives $\text{Vol}_P(\mathcal{F}_Q) \leq \text{Vol}_P(\mathcal{R}_B(D))$. Taking logarithms preserves the inequality, yielding $S_Q(D, B) \leq S(D, B)$. $\square$

Each independent conserved quantity reduces the dimension of $\mathcal{F}_Q$ by one (Proposition 6.5), so the entropy reduction is not merely bounded but progressive: adding conserved quantities strictly decreases S_Q . The semantic connection ∇ enforces further reduction beyond what position-based conserved quantities alone can achieve, because horizontal trajectories must avoid not just forbidden endpoints but forbidden *directions* at every step. In RSVP terms [?], $S(D, B)$ is the semantic entropy of the document: not a measure of disorder but of the volume of admissible futures. The Budget Allocation Principle is the optimal strategy for allocating that volume across semantic dimensions.

8 Profile-Dependent Metric Tensors and Domain Rigidity

Different discourse communities define semantic equivalence differently. An academic paraphrase may freely substitute Latinate for Germanic vocabulary without semantic loss; a legal paraphrase of the same sentence may not, because the choice between *shall* and *must* carries distinct deontic force. These differences reflect genuinely different metric structures on $\mathcal{M}$ and different connection structures on the bundle.

Definition 8.1 (Profile geometry). A *profile geometry* for community P is a triple $(g_P, \nabla_P, \Omega_P)$ where g_P is a Riemannian metric on $\mathcal{M}$ (encoded as a matrix $M_P \in \mathbb{R}^{d \times d}$, $M_P \succ 0$), ∇_P is a semantic connection compatible with g_P , and $\Omega_P = d\nabla_P + \nabla_P \wedge \nabla_P$ is the curvature of that connection. The *profile distance* is $d_P(a, b) = \sqrt{(a - b)^\top M_P (a - b)}$.

Example 8.2 (Academic metric). Let $\hat{u}_{\text{form}}$ be the unit vector from casual to formal register in embedding space. Then $M_{\text{acad}} = I + (w_{\text{form}} - 1)\hat{u}_{\text{form}}\hat{u}_{\text{form}}^\top$, $w_{\text{form}} \gg 1$, makes register-shifting moves expensive under d_{acad} .

Example 8.3 (Legal metric). Under a legal profile, the deontic modality axis (obligation, permission, prohibition) carries amplified weight in M_{legal} . The connection ∇_{legal} forbids horizontal trajectories that cross the obligation–permission boundary, so substitutions altering deontic force are inadmissible regardless of their embedding distance.

Proposition 8.4 (Profile independence of the perturbation algorithm). *The perturbation engine solving (1) is profile-independent: it calls $d_P(\cdot, \cdot)$ and ∇_P as oracles and runs identically for all profiles. Changing profile changes the geometry of the search without altering the algorithm.*

8.1 Metric Learning Interpretation

The profile metric M_P is a *learned Mahalanobis metric* [3]. Given pairs (D_i, D_j) labeled equivalent under profile P by domain experts, M_P is learned by minimizing

$$\sum_{\text{equiv}} d_P(\phi(D_i), \phi(D_j))^2 + \lambda \sum_{\text{distinct}} \max(0, \alpha - d_P(\phi(D_i), \phi(D_j)))$$

subject to $M_P \succeq 0$. This connects profile metrics directly to the metric-learning literature [18]. The connection ∇_P requires richer supervision: pairs of documents together with *trajectories* between them, labeled as admissible or inadmissible by domain experts, are needed to learn the horizontal distribution.

8.2 Curvature and Domain Rigidity

Definition 8.5 (Domain rigidity). Let Ω_P denote the curvature of the semantic connection under profile P and $\kappa_{\text{max}, P}$ the maximum admissible trajectory curvature. The *rigidity* of discourse community P is

$$R(P) = \frac{\|\Omega_P\|}{\kappa_{\text{max}, P}}.$$

High rigidity characterizes domains where the semantic connection is highly curved (parallel transport is strongly path-dependent) and the discourse community tolerates little trajectory curvature. Low rigidity characterizes domains where the connection is flatter and larger loops are permitted without identity failure.

Example 8.6 (Rigidity across profiles). Legal, mathematical, and formal-logical language are expected to exhibit high rigidity: $\|\Omega_{\text{legal}}\| \gg \|\Omega_{\text{literary}}\|$ while $\kappa_{\text{max,legal}} \ll \kappa_{\text{max,literary}}$. In a legal document, the loop *obligation* $\rightarrow$ *permission* $\rightarrow$ *exception* $\rightarrow$ *obligation* may accumulate holonomy: the document returns lexically close to its origin but legally altered. In literary prose, an analogous loop through tonal registers may be perfectly admissible.

The rigidity measure gives a practical implementation rule: a perturbation engine should use profile-dependent gauges $(g_P, \nabla_P, \Omega_P)$ rather than a single universal semantic budget. For legal text, most admissible motion is horizontal and narrow. For literary text, the horizontal distribution is wider and the system can tolerate larger loops without identity failure.

Remark 8.7 (Holonomy as semantic drift). In gauge theory, non-trivial holonomy (a loop in the base space that fails to close in the total space) signals the presence of curvature. In the semantic bundle, holonomy corresponds to meaning drift accumulated along a closed trajectory: a sequence of perturbations that returns to the same embedding position but leaves the document in a different semantic orbit. High-curvature connections ($\|\Omega_P\|$ large) make holonomy more likely and more consequential. Rigidity $R(P)$ measures the ratio of how much holonomy the connection can accumulate to how much the discourse community will tolerate.

9 Multi-Scale Perturbation Operators

Each operator scale draws from its own budget axis and must satisfy both the metric-based budget constraint and the connection-based horizontality condition.

9.1 Lexical Operators

Synonym substitution uses WordNet [19] with frequency-weighted selection. The weight exponent $w_i = \text{freq}(s_i)^\alpha$, $\alpha = 1 - 0.85t$, interpolates between frequency-dominated selection at $t = 0$ and near-uniform sampling at $t = 1$. The per-token replacement probability $p(t) = 0.01 + 0.08t$ controls substitution frequency independently of synonym pool width. At $t \geq 0.5$, an LLM backend is invoked first; WordNet serves as fallback. All proposed substitutions are checked against the conservation constraint and the connection’s horizontality condition before commitment.

9.2 Syntactic Operators

Syntactic operators restructure sentences: active–passive conversion, clause re-ordering, appositive insertion. Syntactic operators incur higher semantic cost ($\Delta \approx 0.008\text{--}0.020$) and are particularly susceptible to connection violations: the documents *A caused B* and *B caused A* differ syntactically in the same way as *A followed B* and *B followed A*, but only the latter is a safe paraphrase under causal graph conservation. All syntactic proposals are submitted to the fiber explorer for both metric and connection checking before commitment.

9.3 Semantic Operators

LLM paraphrase operators rewrite at the sentence or paragraph level ($\Delta \approx 0.015\text{--}0.040$ per sentence). They are invoked only when remaining semantic budget $B_p - \text{spent}_p$ exceeds a threshold. The LLM system prompt encodes profile information, directing the model toward register-appropriate outputs; the prompt also includes protected terms and polarity constraints, partially encoding the connection structure in natural language.

10 Implementation Architecture

The DocPERTURB prototype implements the framework in approximately 3,000 lines of Python and TypeScript across five architectural layers.

The *embedding layer* wraps a sentence-transformer model [16] and provides sentence-level cached embeddings. For a document with 200 candidate substitutions, budget checking without caching requires 200 forward passes; sentence-level caching reduces this to $n_{\text{sentences}}$ passes by recomputing only the modified sentence.

The *semantic budget tracker* maintains the trajectory $D_0 \rightarrow D_k$, computing total displacement and per-step distance for every proposed mutation. The completion field $\gamma(c, t)$ is tracked incrementally, updating after each committed mutation, and budget allocation follows Theorem 7.5.

The *constraint and connection engine* implements the conservation constraint (Definition 6.2), the gauge-equivalence check (Definition 6.6), and a partial implementation of the connection horizontality condition (negation polarity, simple causal patterns via dependency parsing). Full graph-level conservation is reserved for future work.

The *perturbation operators* implement lexical, syntactic, and semantic mutations, each submitting proposals to a unified `FiberExplorer.admit()` interface that checks metric budget, connection horizontality, and conservation before commitment.

The *search loop* performs greedy optimization of (1), allocating budget to components according to the colored noise principle. Future work will explore beam search and simulated annealing within the budget constraint [5].

11 Experimental Evaluation

We outline experimental protocols; full empirical results are reserved for a follow-up study with access to human annotators and standardized benchmark corpora.

Semantic fidelity versus temperature. For each temperature $t \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ and each of the six profiles, we generate 100 variants of 50 source documents across diverse genres. We measure embedding cosine distance, human judgments on a 5-point scale, and preservation of named entities and key claims.

Colored noise versus white noise. We compare uniform budget allocation against the completion-weighted allocation of Theorem 7.5, measuring semantic diversity (reachable set hull volume), conservation constraint satisfaction, and human naturalness ratings.

Connection violation detection. We measure the rate at which the connection-based horizontality check (negation and causal graph conservation) blocks mutations that pass the metric-based budget check but would alter propositional meaning, evaluating the incremental benefit of connection checking over metric checking alone.

Domain rigidity measurement. For each profile, we empirically estimate $R(P) = \|\Omega_P\| / \kappa_{\max, P}$ by measuring how frequently valid (budget-admissible) perturbations produce meaning changes judged unacceptable by domain experts. We predict $R(\text{legal}) \gg R(\text{creative})$.

Retrieval robustness and deduplication benchmarking. Perturbed variants at calibrated budgets test whether dense retrieval systems correctly identify near-duplicates. Budget B provides the ground-truth semantic distance label: pairs within $B = 0.05$ are “equivalent,” between 0.05 and 0.15 “related,” beyond 0.15 “distinct.”

12 Applications

Data augmentation. Perturbations within small budget B preserve class labels while increasing diversity; budget provides a formal guarantee that the label has not been corrupted.

RAG robustness evaluation. Semantic perturbations of queries and source documents at calibrated distances test retrieval system sensitivity to paraphrase.

Plagiarism detection calibration. Perturbations at increasing budgets generate test cases at known semantic distances, identifying the budget threshold at which detection fails.

Adversarial robustness. Selectively violating conservation constraints or connection horizontality generates adversarial examples targeting specific semantic structures.

Theory version control. Any object admitting a semantic embedding—a scientific theory, a software specification, a legal statute—can be tracked under the framework. A theory evolves admissibly if each revision stays within a semantic budget relative to the founding formulation, and conserved quantities ensure core claims are not inadvertently abandoned.

13 Limitations and Open Problems

Metric tensor and connection construction. Profile metrics are currently initialized from hand-specified contrastive pairs; connections are partially implemented. Without learned connections, the horizontal distribution is only approximately enforced via LLM prompting. Learning ∇_P from trajectory-level supervision is the most important open problem.

Computational cost. Each proposed mutation requires an embedding forward pass and, if connection checking is active, a dependency parse and graph comparison. Sentence-level caching and batched mutation proposals mitigate cost; approximate fiber-membership testing via locality-sensitive hashing is a further direction.

Completion field estimation. Theorem 7.5 assumes $\gamma(c, t)$ can be measured. In practice, decomposing a document’s semantic content into well-defined components and estimating completion from embeddings alone is non-trivial. Structured semantic representations (knowledge graphs, frame-semantic parses) would make this reliable.

Non-holonomic language phenomena. The connection formalizes the intuition that certain semantic failures are directional rather than positional. But the full non-holonomic structure of language—scope, presupposition, implicature, discourse coherence—is far richer than the polarity and causality examples developed here. Mapping the full horizontal distribution of natural language remains an open problem in formal semantics.

Embedding faithfulness. The framework rests on the assumption that embedding cosine distance approximates human semantic distance, which is known to fail for negation, presupposition, and other compositional structures. The connection addresses some failures; others require structured representations beyond sentence embeddings.

14 Future Directions

Learning the semantic connection. Tracking how causal graphs, rhetorical trees, and polarity structures are preserved across human-authored paraphrases in large corpora would effectively map the horizontal distribution of the semantic bundle, making the connection data-driven rather than rule-based.

Adaptive budget allocation. Updating $\gamma(c, t)$ after each committed mutation and reallocating budget dynamically would better track the actual exploration frontier than the current static initialization.

Graph-based conservation. Replacing flat embeddings with knowledge-graph or frame-semantic representations would enable conservation at the propositional

level, addressing negation and causality failure modes within the conservation formalism rather than via ad-hoc rules.

Multimodal perturbation. The budget, connection, and conservation framework generalizes to any modality admitting a semantic embedding. Multi-modal documents could be perturbed jointly under a shared semantic budget and a joint connection.

Connection to RSVP and projection formalisms. The fiber exploration framing connects naturally to the RSVP field-theoretic framework [?], in which admissible configurations satisfy a projection condition analogous to fiber membership and entropy measures the volume of admissible future trajectories. The semantic accessibility entropy $S(D, B)$ is, in RSVP terms, the semantic entropy of the document. The Budget Allocation Principle is the optimal strategy for allocating that entropy across semantic dimensions. The connection ∇ and its curvature Ω_P correspond to the field structure of the RSVP plenum, and admissible perturbations correspond to horizontal lifts of semantic trajectories in the same way that admissible physical configurations correspond to field configurations satisfying the RSVP projection conditions. Diffusion sampling (CNS), document mutation (DocPerturb), and RSVP accessibility dynamics are therefore all instances of a single abstract principle: constrained finite-budget perturbation in which energy is routed toward unresolved dimensions rather than distributed uniformly.

15 Conclusion

We have argued that document perturbation is fundamentally a geometric problem. A document is a point on a semantic manifold; the text file is one coordinate representation of an underlying geometric object; admissible paraphrases occupy the same semantic fiber; and controlled augmentation is constrained trajectory planning within that fiber under a budget of semantic distance and a connection that governs admissible transport of meaning.

The framework contributes a complete mathematical narrative: embedding $\rightarrow$ manifold with local linearization $\rightarrow$ tangent decomposition $\rightarrow$ equivalence classes $\rightarrow$ fibers and sections $\rightarrow$ reachable sets $\rightarrow$ budget vectors $\rightarrow$ semantic velocity and curvature $\rightarrow$ semantic connection and horizontal lifts $\rightarrow$ holonomy and accumulated drift $\rightarrow$ conservation as geometry (Noether-style invariants, gauge freedom, invariant sub-manifold) $\rightarrow$ entropy reduction by conservation $\rightarrow$ colored noise allocation (Budget Allocation Principle, accessibility entropy, monotonicity) $\rightarrow$ profile-dependent geometry and domain rigidity.

Each step does genuine mathematical work. The tangent decomposition connects the manifold geometry directly to the operator hierarchy. The semantic connection resolves the non-holonomic failure modes that distance metrics alone cannot detect. The holonomy subsection gives curvature a concrete operational meaning before it reappears in the profile section. Proposition 7.10 makes precise the verbal claim that conservation laws reduce semantic accessibility, providing the formal link between the conservation and entropy threads. The Budget Allocation Principle shows that uniform perturbation is suboptimal. The domain rigidity measure $R(P) =$

$\|\Omega_P\|/\kappa_{\max,P}$ makes precise the intuition that legal and mathematical language tolerate less semantic perturbation than literary language—not because of stricter budgets but because of genuinely different geometric structures on the semantic bundle.

What began as a synonym randomizer has converged on a gauge-theoretic framework for controlled semantic transformation. The synonym replacement machinery is one implementation detail; the framework accommodates any operator that maps documents to documents, admits a semantic cost, and can be checked for horizontal motion under the active profile connection. As embedding models improve, as connection learning from trajectory-level supervision matures, and as structured semantic representations become standard, the architecture is positioned to scale from the proof-of-concept demonstrated here to a robust, application-grade perturbation engine applicable to language, cognition, software evolution, version control, scientific theories, and any other domain whose objects admit a projection onto a semantic manifold.

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